

Given an array of N integer, we have to maximize the sum of $arr[i] * i$, where i is the index of the element ($i = 0, 1, 2, \dots, N$). Write an algorithm based on Greedy technique with a Complexity $O(n \log n)$.

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int compare(const void *a, const void *b)
5  {
6      int x = *(int *)a;
7      int y = *(int *)b;
8
9      return x - y;
10 }
11
12 int main()
13 {
14     int n;
15     scanf("%d", &n);
16
17     int a[n];
18
19     for (int i = 0; i < n; i++)
20     {
21         scanf("%d", &a[i]);
22     }
23
24     qsort(a, n, sizeof(int), compare);
25
26     long long sum = 0;
27
28     for (int i = 0; i < n; i++)
29     {
30         sum = sum + (long long)a[i] * i;
31     }
32
33     printf("%lld", sum);
34
35     return 0;
36 }
```

	Input	Expected	Got	
✓	5 2 5 3 4 0	40	40	✓
✓	10 2 2 2 4 4 3 3 5 5 5	191	191	✓
✓	2 45 3	45	45	✓

Passed all tests! ✓